CCCA Cell-State Analysis

Gulati et al. 2020 · Breast Cancer

SmartSeq2 · 861 malignant cells (8 samples) → 587 analysed (4 samples ≥100) · non-branching geometry

GO / NO-GO GATES

Gate 1 — batch mixing & continuity — GO

Four samples (≥100 malignant cells) intermix per-neighbourhood on one continuous manifold — iLISI rises 2.77→3.56 of 4 after Harmony (θ=5), one connected component, no islands removed. Read on the regenerated embedding, not a scalar alone.

Gate 2 — embryonic module geography — GO

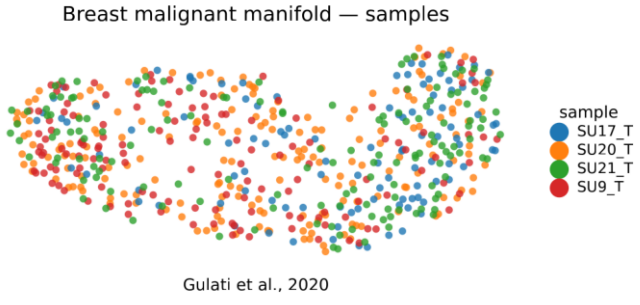
A single (Gast/pSom)→Crest developmental axis. Gast/pSom do NOT diverge into cross-repressing arms (raw r=−0.05), while both oppose Crest (Gast−Crest r=−0.50, pSom−Crest r=−0.60). Crest is the inverse terminal readout (rises along pseudotime ρ=+0.93; NC-high decile at pt 1.00; joint pSom&Crest enrichment 0.31<1). Non-branching case → GO.

Gate 3 — hub cross-repression — NA

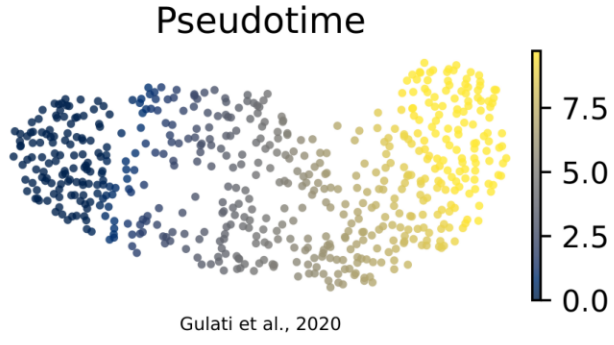
No bifurcation (curve: 2 tips / 0 forks), so there is no Mixed hub to test for Gast↔pSom anti-correlation. Branching-only → NA; 04_correlation / 05_modules not produced.

STEP FIGURES

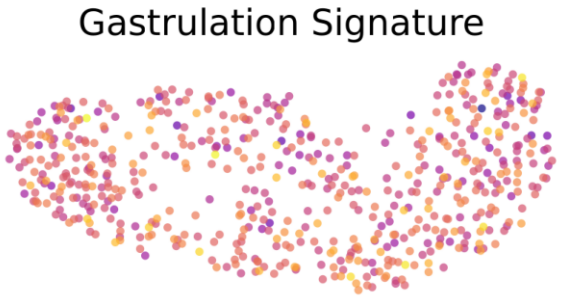
01 · Samples (integration)



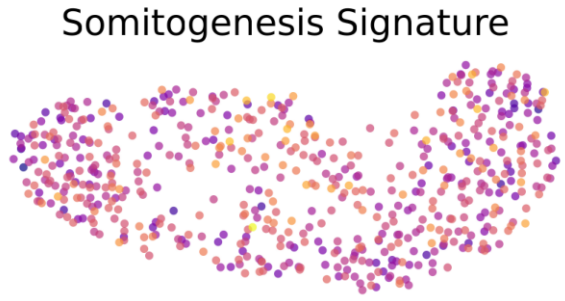
02 · Pseudotime (curve)



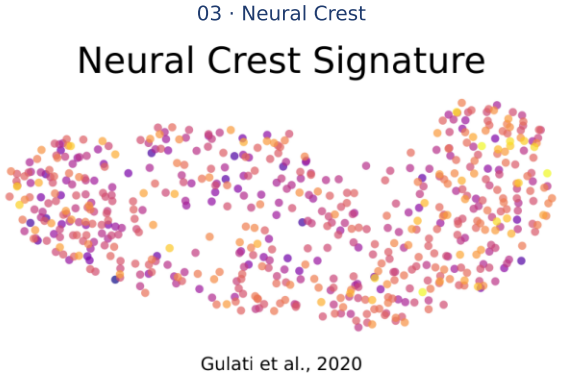
03 · Gastrulation



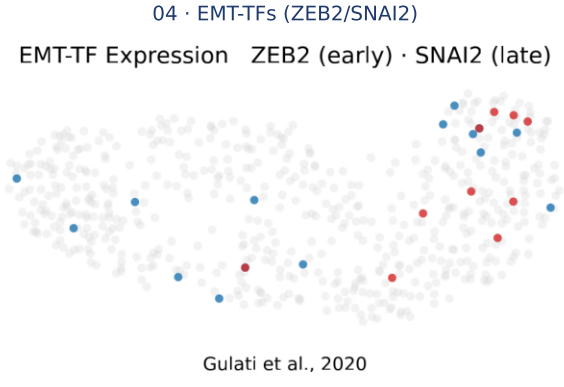
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs (ZEB2/SNAI2)



PARAMETER CHOICES & RATIONALE

Malignant / threshold / analysed	861 → floor ≥100 (counts smooth, no dominant cliff) → 4 samples / 587 cells; 0 islands removed
Expression / integration	SmartSeq2 TPM (log1p); overdisp=98; n_pcs=20; Harmony(sample) θ=5; k=30; UMAP spread=1.3; seed=1
Trajectory / geometry	scFates curve (ElPiGraph) Nodes=20, epg_lambda=0.3, epg_mu=100; 2 tips / 0 forks; fold self-approach=2.89
Root	curve tip 0 (Gast/pSom-high, Crest-low developmental origin); Crest terminal; axis cell-cycle-free (pt vs prolif ρ=−0.01)
Subtype / annotation	single low-cycling Body (no proliferative compartment scoped out); publication cell_type = Malignant
EMT-TFs	ZEB2 (root-end) / SNAI2 (terminus) from 10-TF panel incl. FOXC/PRRX; detection-gated; VIM excluded (not a TF)

VERDICT

Pattern reproduces (non-branching, Linear-pSom/Crest): a single Gastrulation/Somitogenesis → Neural-Crest developmental axis with Crest as the inverse terminal readout; no Gast↔pSom bifurcation (Gate 3 NA).